Department of Computer Science

CS5701 Quantitative Data Analysis

Postgraduate Study Guide for 2026/27

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MODULE DETAILS

Module Leader	Dr Isabel Sassoon
Department	Computer Science
Credits	n.a. (study block only)
Other staff	Dr Sarath Dantu Dr Leila Ghanbar
Contact and private study time	Lectures 9 hours Labs 18 hours Coursework Study 50 hours General Study 73 hours Total 150 hours
Assessment	For the MSc Data Science and Analytics and MSc Artificial Intelligence programmes, assessment is covered by two separate assessment blocks: CS5801 Quantitative Data Analysis CS5802: Critical Analysis of Modern Data.

ACCESS TO SUPPORT MATERIAL AND ADDITIONAL INFORMATION

Most teaching, learning and support material is provided electronically via the University's Brightspace system. Note that the details provided in this study guide are based on the formal module syllabus for this module which sets out the agreed content, learning outcomes, assessment and teaching methods. Module syllabus and scheme of studies documents for your programme of study can be found by on the University's Quality Assurance web pages.

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AIMS

The aim of this module is to develop knowledge and skills of the quantitative data analysis methods that underpin data science. The course content covers a practical understanding of core statistical methods in data science application and research (e.g., exploratory data approaches, univariate and multivariate methods, regression and ANOVA). A focus is also placed on learning to evaluate the strengths and weaknesses of methods alongside an understanding of how and when to use or combine methods, interpret the output and present findings in a clear, concise and understandable manner.

By the end of the course, students will be able to appreciate the role of statistical methods and to develop skills in presenting quantitative data using appropriate displays, tables and summaries. Students will also learn about the role of statistical methods in quantifying variation, setting confidence limits, and testing hypotheses, with applications in the real world including in the context of AI.

The study block will use the R-ecosystem as its primary toolset, the study block will provide hands-on and practical lab time where the quantitative data analysis methods explored will be applied on different data sets and analyses. R is widely used for statistical analysis and data science applications.

LEARNING OUTCOMES

Whatever module or programme of study you are studying for at Brunel University London, there are learning outcomes (LO) that you must meet/achieve in order to be awarded the credits which comprise the module and programme of study. In order to get a pass grade (C- or above) in this module, you must meet these learning outcomes below, that is, you must demonstrate ability to:

LO1: Develop understanding of statistical data analysis methods and their relevant use in data science application and research.

LO2: Learn to critically evaluate the strengths and the weaknesses of statistical data analysis methods alongside an understanding of how and when to use or combine methods.

METHOD OF TEACHING

This module is intended to be both theoretical and practical in its nature.
For each week of the module there will be:
1. A scheduled classroom lecture
2. A scheduled lab session that will consist of two parts: The first part will be a guided session which will be followed by a supervised independent study activity.

The schedule is outlined in the next section.

LECTURE PROGRAMME

This module has been scheduled to run during Semester 1 and the lectures are scheduled as follows:

Semester 1

Week	Lecture Topic	Lecturer
1	Exploring Data	IS
2	Inferential Statistics – Part 1	IS
3	Inferential Statistics – Part 2	IS
4	Count Data and Table Analysis	IS
5	Linear Regression	IS
6	READING WEEK	
7	Analysis of Variance (ANOVA)	IS
8	Multiple Regression	IS
9	Generalised Linear Models (GLM) and Logistic Regression	IS
10	QDA in the wild	IS
11	Revision Lecture	IS

Labs are scheduled as follows:

Semester 1

Week	Date	Lab Topic
1	30/09/2026	Exploring data in RStudio
2	7/10/2026	Inferential Statistics in RStudio
3	14/10/2026	Inferential Statistics in RStudio
4	21/10/2026	Table Analysis in RStudio
5	28/10/2026	Linear Regression in R Studio
6		
7	11/11/2026	ANOVA in R Studio
8	18/11/2026	Multiple Regression in R Studio
9	25/11/2026	LR in R Studio

Week	Date	Lab Topic
10	02/12/2026	QDA in the wild
11	11/12/2026	Revision Drop in Lab

Please note: Week 6 in Semester 1 and week 23 and 26-28 in Semester 2 are weeks during which there are no scheduled lectures, labs, seminars or tutorials. Weeks 15-17 are the Winter Examination Period. Week 18 is the first week of teaching after the Christmas break.

READING LIST

Core reading list

Crawley, M.J., 2015. Statistics: an introduction using RJ Wiley, England. Available from the Library and as a digital download.

Supplementary Reading

Jones, E., Harden, S. and Crawley, M.J., 2022. The R book. John Wiley & Sons. Available https://onlinelibrary-wiley-com.ezproxy.brunel.ac.uk/doi/book/10.1002/9781119634461

Chatfield, C., 1995. Problem solving: a statistician's guide. CRC Press. Selected chapters.

Matloff, N., 2019. Probability and statistics for data science: Math+ R+ data. CRC Press. Available to read online: https://ebookcentral.proquest.com/lib/brunelu/detail.action?docID=5795811

R in Action: Data Analysis and Graphics with R: https://learning.oreilly.com/library/view/r-in-action/9781617291388/

R ggplot resource: http://r-statistics.co/ggplot2-Tutorial-With-R.html

Campbell, M., 2020. Learn RStudio IDE. New York: Apress. Downloadable pdf https://link-springer-com.ezproxy.brunel.ac.uk/book/10.1007%2F978-1-4842-4511-8

Supplementary Reading

Please note that there is a wealth of material, much of it freely available on the web or in the library. The above are suggestions but you are encouraged to search for and make use of other sources.

It is important that you learn to become self-reliant and able to access and assimilate material for yourself. Many of the topics necessary for this project will not be covered by lectures. Therefore, you will need to work through the exercises and guidance material provided in labs, via Brightspace or from your own investigations.

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ASSESSMENT

This study block is offered on the following programmes:

- MSc Data Science and Analytics
- MSc Artificial Intelligence.

The MSc Data Science and Analytics and MSc Artificial Intelligence programmes include integrated assessments. The relevant assessment blocks are:

- CS5801 Modern Quantitative Data Analysis
- CS5802 Critical Analysis of Modern Data

CS5801 and CS5802 are the two integrated assessment blocks for this study block (CS5701) and CS5702 Modern Data study block. Details on the assessment frameworks will be presented during the first lecture.

LATE COURSEWORK

The clear expectation is that you will submit your coursework by the submission deadline stated in the study guide. In line with the University's Coursework Submission Policy (approved in April 2025), coursework submitted up to 48 hours late will be accepted but the penalties set out in the 'Late Submission' section of the policy will be applied. Work submitted over 48 hours after the stated deadline will automatically be graded NS (non-submission).

Please refer to the Computer Science student information pages and the Coursework Submission Procedure pages for information on submitting late work, penalties applied and procedures in the case of Exceptional Circumstances.

DELIVERABLES AND FEEDBACK – IMPORTANT DATES

You should prepare and submit all coursework according to the instructions in the relevant assessment brief. You should make sure that you are fully aware of the University's policy on plagiarism and collusion. You should also be aware that you *cannot* later claim that you did not know the rules and regulations as you must make yourself familiar with them. If you cannot complete any work on time, you should look at the Department's instructions on what to do. The Department policy is that all coursework must be submitted electronically via the WISEflow system. Please navigate to the Brightspace pages for this module for further details. You will get feedback on your performance via WISEflow for this module. If do not receive your feedback by the given date, you should first contact the module leader. If it proves necessary, you should also contact the Director of Postgraduate Studies.

The important dates:

For assessment brief availability, submission and feedback deadlines, please consult the assessment briefs of the relevant assessment blocks CS5801 and CS5802.

Note: all deadlines are at 11.00 a.m. UK time on the stated date (e.g., if the deadline is 1 February, it means the deadline is 11 a.m. UK time on 1 February)

Expectations of Artificial intelligence Use

The University has general guidance on using artificial intelligence in your studies.

Generative AI and Large language models (LLMs) such as chatGPT and Claude all have the potential to answer questions on code and material relevant to this module. Sometimes the answers are of very high standard with helpful accompanying explanations, sometimes they are wrong. A difficulty for a learner is to know the difference. If you do not learn how to do quantitative data analysis yourself then it will be difficult for you to know the difference between a correct or incorrect answer.

Employability and Career Skills

This module develops key employability skills in data analysis, statistical reasoning, problem solving and interpretation of analysis outputs. Through practical work in RStudio, students learn how to analyse data, evaluate results and communicate findings effectively, providing a strong foundation for careers in data science, artificial intelligence, research and other data-driven professions. The teaching and assessment are constructively aligned, with one of the two assessment blocks designed as an authentic assessment that reflects a real-world data analysis project, enabling students to practice applying statistical techniques, interpret findings and communicate conclusions in a professional context.

Additional Vital Information

Information relating to all aspects of your studies, including procedures of how to inform us of problems you are facing with your studies, how to apply for an extension for your coursework submission, plagiarism and types of other academic misconduct, and other important matters, are available on the University's intranet. The Department assumes that you familiarise yourself with this information, so you will need to look at the relevant intranet pages carefully at various times throughout your studies. The Department also operates within the rules and regulations of the University more generally, and you should also look at what are known as 'Senate Regulations' on the University's web pages. These policies and procedures might change from one academic year to another, and it is in your interest to review them on an on-going basis.